When More Observation Hurts: Redundancy and Robust Clarification in Ask-or-Act Under Instruction Ambiguity

Zhu Yufan

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Abstract. A cooperative assistant can infer a principal's goal by passively observing actions or actively asking costly questions. No prior benchmark combines both channels; we build the first and uncover three findings. (1) The channels are used *sub-additively* (I=−26±3 pp): a cost-aware policy extracts overlapping information from each source. (2) Assistability is *non-monotonic* in rationality: higher β can *hurt* because action aliasing hides intent (92%→84% at K=3, β:8→16). (3) Under model mismatch, passive observation *poisons* the posterior (−2.2 pp/step vs. −0.8 matched); three decision-layer fixes all fail, but surprise-tempered updates (AskOrAct-ST) recover +1–+5 pp with zero matched-model harm.

1 Introduction

A central challenge in cooperative assistance is deciding when a helper should ask for clarification rather than act. Acting immediately risks committing to the wrong goal, whereas asking too often wastes time and may frustrate the principal. The trade-off depends on the informativeness of available questions, the cost of a wrong action, and what the principal's behaviour already reveals.

We study this question by constructing an attribute-structured, post-parsing clarification benchmark in which the candidate goal set is known, question semantics are fixed, and the principal model is given. The benchmark's value is controlled diagnosis, not ecological realism. It includes a cooperative principal whose actions are observable for passive inference via inverse planning, and a question bank the assistant may use at explicit cost for active clarification. The contributions are threefold, comprising a reproducible two-room gridworld with templated instruction ambiguity and a Boltzmann principal model, a systematic comparison of six ask-or-act policies across 19,800 episodes, and an empirical characterisation of how the two channels interact. Our contribution is not a new ask policy, but, to our knowledge, the first controlled characterisation of how the value of active clarification changes as passive action evidence weakens, with the empirical success–question-cost frontier shifting substantially with principal rationality.

Why a controlled benchmark? Prior work studies each channel in isolation: CLIPS [12] performs inverse planning without asking, while ATD [8], SAGE [11], and Grand et al. [4] study clarification without action-based inference. To our knowledge, no prior benchmark combines both channels with controlled ambiguity sweeps; Table 2 in Appendix A shows the comparison. The gap is the ability to study how passive and active evidence *interact* under a shared objective.

Three Takeaways. (1) The channels are used *sub-additively* (I=−26 pp at K=3): AskOrAct extracts overlapping information; clarification matters most when passive inference is structurally weak. (2) Assistability is *non-monotonic* in β: extreme rationality collapses action diversity and hides intent, so NeverAsk success falls at high β. (3) Under model mismatch, forced observation costs −2.2 pp/step (2.8× the matched rate), explaining why passive-aware decision-layer heuristics fail. The benchmark is intended as a reusable testbed for future ask-or-act methods.

2 Related Work

Inverse planning infers goals from behaviour [1, 9]; CIRL [5] formalises cooperative teams; CLIPS [12] adds language evidence—none provides an ask-or-act decision. Shared autonomy [6] and HRI clarification [2, 13] study one channel at a time. Goal recognition design [3, 7] and PAGR [14] target recognition, not task completion with costly typed questions. ATD [8], SAGE [11], and Grand et al. [4] study clarification without observing principal actions, precluding measurement of channel interaction. Our benchmark is the first to control both channels simultaneously under a shared success–cost objective.

3 Method

3.1 Environment

We use a 9×9 two-room gridworld (Figure 3, Appendix E) with M=6 object instances per episode. Room 0 and Room 1 are separated by a mid-wall with a single doorway. Each episode samples 6 objects from types {key, gem, coin} and colours {red, blue, green}; exactly K match the ambiguous instruction (candidate goals) and $M-K$ are distractors.

A principal has a hidden goal object. An instruction template maps the goal to a surface form shared by K candidate goals (ambiguity level K). The principal takes approximately rational actions towards its goal, sampled from a Boltzmann policy with rationality β and ε-greedy noise: $\pi_{\beta,\varepsilon}(a \mid s, g) = (1-\varepsilon)\,\mathrm{softmax}_\beta\big(V(s,\cdot,g)\big)_a + \varepsilon/|A|$, where $V(s,a,g) = -d^*(\mathrm{succ}(s,a),\, g)$ is the negated shortest-path distance from the successor of a to g.

The assistant observes the state, instruction, and all principal

actions, but not the true goal. An episode ends when the assistant picks an object, which counts as success if correct, or when the step budget $T_{\max}$=200 is exhausted. The question bank Q contains three types of questions addressing the goal's colour, room, and object type, each with a factual answer drawn from a per-type noisy channel. When the assistant decides to ask, it remains at its current position. When it decides to act, it takes one step toward the MAP-goal object via A* navigation and then re-evaluates.

Full hyperparameters are in Appendix E, Table 5; the cooperative setup is illustrated in Appendix E, Figure 4.

3.2 Bayesian Posterior

Per-step protocol. The assistant observes the principal's action, updates b_t, then asks or acts; if asking it updates b_t again from the answer, if acting it moves one A* step toward the MAP goal.

The assistant maintains a posterior $b_t(g)$ over candidate goals $\mathcal{G}$. Initialisation uses the instruction likelihood under each goal. After observing principal action a_t at state s_t:

$$b_{t+1}(g) \propto b_t(g) \cdot \pi_{\hat{\beta},\hat{\varepsilon}}(a_t \mid s_t, g),$$

where $\hat{\beta}, \hat{\varepsilon}$ are the assistant's assumed principal parameters, matched to the true values unless otherwise noted in the mismatch analysis.

When a question q is asked and answered α, the posterior is updated by the answer likelihood:

$$b(g) \propto b(g) \cdot P(\alpha \mid q, g, \text{noise}),$$

where the answer noise varies by type, with color at 0.05, room at 0.10, and object at 0.20.

3.3 Ask-or-Act Policy

AskOrAct uses a Bayes-risk ask gate to decide *whether* to ask, paired with a separate MAP+A* controller for motion execution. The ask gate compares the expected surrogate team cost, defined as navigation steps plus per-question penalty c_q, of asking versus acting under the current posterior b_t. The policy is not a unified Bayes-risk controller since the ask decision uses expected cost while the act branch greedily follows A* toward the MAP goal. It serves as an analytically computable reference for locating one operating point on the empirical success–question-cost frontier, not as a proposed optimal policy.

Formally, at step t, the assistant computes:

$$\text{CostAct} = \sum_g b_t(g) \cdot d^*(s_t, g),$$

the expected obstacle-aware shortest-path distance computed via A* to each candidate goal under the current posterior, and

$$\text{CostAsk}(q) = \underbrace{1}_{\text{step}} + \underbrace{c_q}_{\text{question}} + \mathbb{E}_\alpha\big[\text{CostAct}(b_{t,q,\alpha})\big],$$

where asking incurs both one timestep of delay costing 1 and an additional penalty $c_q = 0.5$ for the principal's effort, and the expectation integrates over answers α after updating to $b_{t,q,\alpha}$. This one-step myopic lookahead does not plan over multi-step ask sequences.

Decision rule. The policy selects $\arg\min_q \text{CostAsk}(q)$ when $\min_q \text{CostAsk}(q) < \text{CostAct}$, and acts otherwise. Because the surrogate cost C does not penalise wrong picks, this comparison minimises effort under the current posterior but does not account for failure risk.

Heuristic guards. Three constraints are imposed for robustness and are not part of the decision-theoretic derivation: (i) entropy gate $H(b_t) \leq 0.3$ provides a fast exit; (ii) asking is disallowed after step 6; (iii) at most $Q_{\max}$=3 questions per episode. All thresholds are hand-set; the ablation in Section 5 shows c_q dominates (Appendix D). The full decision flow is illustrated in Appendix F, Figure 5.

3.4 Surprise-Tempered Updates (AskOrAct-ST)

Under model mismatch, standard Bayesian updates can poison the posterior as shown in Section 5. We address this with a surprise-triggered likelihood tempering that modifies the passive update without changing the ask-or-act policy:

$$b_{t+1}(g) \propto b_t(g) \cdot \pi(a_t \mid s_t, g, \hat{\beta})^{\omega_t},$$

where $\omega_t \in (0, 1]$ controls how much the passive observation is trusted. The tempering weight is driven by a CUSUM-style surprise accumulator:

$$S_t = \max\big(0,\ \gamma\, S_{t-1} + \big(-\log p_t(a_t) - \mu\big)\big), \quad \omega_t = \frac{1}{1 + S_t/\tau},$$

where $p_t(a_t) = \sum_g b_t(g)\, \pi(a_t \mid s_t, g, \hat{\beta})$ is the predictive action probability, μ=1.5 is the matched-model baseline surprise, γ=0.9 is the forgetting rate, and τ=3.0 is the tempering scale. Under a matched model, $S_t \approx 0$ so $\omega_t \approx 1$, recovering standard Bayesian updating. Under mismatch, S_t grows and ω_t shrinks, downweighting unreliable passive evidence.

This intervenes at the *inference* layer rather than the decision layer, and the ask-or-act gate remains unchanged. Three prior attempts to improve the decision layer all failed as described in Section 5, motivating an inference-layer intervention instead.

3.5 Baselines

All baselines share the same act branch (A* toward MAP goal) and differ only in their ask-gating condition; formal conditions appear in Appendix F, Table 7. **NeverAsk** never asks. **AlwaysAsk** asks the highest-IG question whenever the entropy and budget gates permit. **InfoGainAsk** adds an information-gain threshold gate. **RandomAsk** uses the same gates as InfoGainAsk but picks a question uniformly at random. **EasyInfoGainAsk** ($Q_{\max}$=1, no entropy gate) and **POMCPPlanner** (POMCP [10], 250 iterations, horizon 8) are side diagnostics evaluated only at K=2.

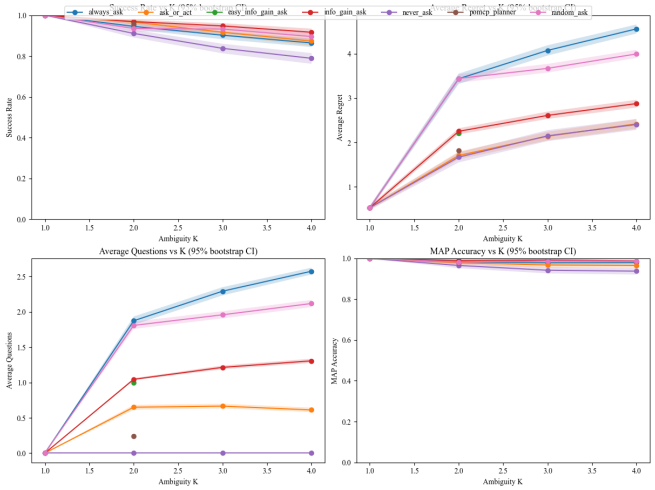

Figure 1: Success rate, regret, and questions asked by ambiguity K (averaged over ε, β; error bars: 95% bootstrap CI). Policies diverge at $K \geq 3$; InfoGainAsk trades more questions for higher success.

4 Experimental Setup

Sweep. The main evaluation covers $K \in \{1, 2, 3, 4\}$, $\varepsilon \in \{0, 0.05, 0.1\}$, $\beta \in \{1, 2, 4\}$, 5 replicate seeds × 20 episodes per condition = 100 episodes per cell. The 5 core policies are evaluated in every cell, and *EasyInfoGainAsk* and *POMCP-Planner* are additionally evaluated at K=2. This yields a total of 19,800 episodes.

Metrics. The two primary axes are *success rate*, the fraction of episodes where the assistant picks the true goal, and *questions asked*, averaged per episode. Together these define the empirical success–question-cost frontier. We also report the *surrogate team cost* C=steps+c_q·|questions|, the effort scalar AskOrAct minimises. Note that C does not penalise wrong picks, as success rate captures that dimension. *Regret* $= C - C^*_{\text{oracle}}$ measures excess effort relative to the oracle's shortest-path cost C^*_{oracle}. Because C does not penalise wrong picks, regret is a secondary diagnostic and success rate is the primary outcome.

Statistics. Key claims use *paired seed-level deltas* (mean ±1 s.e. over 5 seeds); figures show 95% bootstrap CIs (B=1000). Full per-(K, ε, β) breakdowns appear in Appendix C.

5 Results

5.1 Main Sweep

Figure 1 shows results averaged across all ε and β conditions. At K=1, where there is no ambiguity, all policies perform identically with 100% success and zero questions. As ambiguity rises to $K \geq 3$, policies diverge.

Caveat: K is not a pure manipulation. Template feasibility changes with K (e.g. T2 is excluded at $K \geq 2$). A within-template T3 control restricts to the color-only template across $K \in \{2, 3, 4\}$; AskOrAct–NeverAsk deltas of −2, +3, +7 pp confirm the advantage grows with K (Appendix C, Table 3).

Table 1 summarises the key aggregate results, and full per-(K, ε, β) results appear in Appendix C.

Table 1: Mean results averaged over all (ε, β) conditions. C = surrogate team cost (steps +c_q· questions; lower is better). **Bold**: best success; uline: lowest C (secondary).

K	Policy	Success	$C\downarrow$	Regret	Q/ep
2	InfoGainAsk	**96.9%**	8.78	2.25	1.05
	AskOrAct	96.7%	<u>8.04</u>	1.71	0.65
	AlwaysAsk	94.6%	10.31	3.44	1.87
	NeverAsk	91.1%	8.20	1.67	0.00
3	InfoGainAsk	**94.8%**	9.82	2.61	1.21
	AskOrAct	91.7%	<u>8.81</u>	2.14	0.67
	AlwaysAsk	90.2%	11.87	4.08	2.29
	NeverAsk	83.8%	9.23	2.15	0.00
4	InfoGainAsk	**91.7%**	10.54	2.88	1.31
	AskOrAct	87.3%	<u>9.43</u>	2.42	0.61
	AlwaysAsk	86.4%	13.08	4.56	2.58
	NeverAsk	78.9%	9.88	2.40	0.00

The frontier: InfoGainAsk vs. AskOrAct. *InfoGainAsk* attains +3.1–+4.4 pp higher success at K=3, 4 using ~2× more questions. AskOrAct achieves the lowest surrogate cost C at every K, and neither policy is universally preferable without an application-specific exchange rate.

Gain from asking: AskOrAct vs. NeverAsk. At K=3 the gain is +7.9 ± 1.2 pp, with mean ± s.e. over 5 seeds and a 95% CI of [5.0, 10.8]. At K=4 the gain is +8.4 ± 1.4 pp with a 95% CI of [5.1, 12.1]. The effect is positive for every seed with only 0.61–0.67 questions per episode. *AlwaysAsk* asks 2.3–2.6× more but achieves lower or similar success, confirming over-asking is costly.

5.2 Robustness

Answer noise (exploratory). An independently-sampled sweep varies answer noise from 0 to 0.2. Because episodes are not seed-matched across levels, this analysis is exploratory as detailed in Appendix G.

β-mismatch. With $\hat{\beta}$=2 fixed and true $\beta \in \{1, 2, 4\}$, the hardest case of β=1 with a near-random principal still yields +13.5 pp for AskOrAct over NeverAsk.

Scale-K. Extending to $K \in \{5, 6\}$ shows that the asking advantage grows. *AskOrAct* retains meaningful success while *NeverAsk* falls substantially, consistent with VoI being most valuable under high ambiguity.

Non-monotonic assistability in principal rationality. Figure 2 sweeps β from 0.5 to 16 with matched models. At K=3, NeverAsk success peaks at β=8 with 92% and then *drops* to 84% at β=16 because extreme rationality collapses action diversity among aliased goals, hiding the principal's intent. AskOrAct detects this and responds by increasing its Q/ep from 0.52 at β=8 to 0.60 at β=16, maintaining 92%. At K=4 the effect is weaker, and we report it as regime-dependent rather than universal.

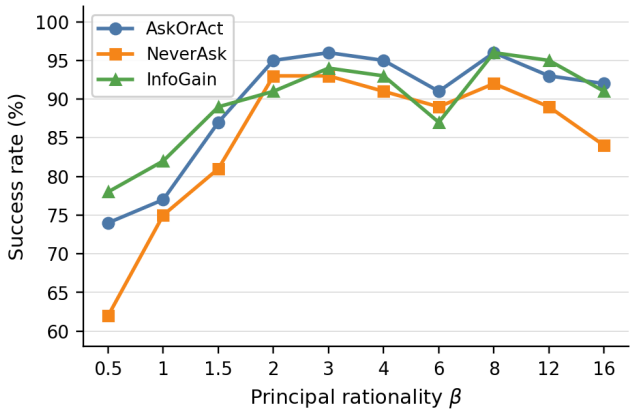

Figure 2: Success vs. β at K=3. NeverAsk drops at high β (action aliasing); AskOrAct compensates. K=4 shows a weaker effect (not shown).

Path overlap determines ask-worthiness. When candidate goals share the same optimal first action (full aliasing), NeverAsk collapses to 64.5% while AskOrAct maintains 87.1% (+22.6 pp). When goals diverge spatially, both policies match at 100%. Episode-level stratification by overlap appears in Appendix G, Table 12.

More observation can hurt under mismatch. Forcing h pre-decision observation steps at K=4: under matched models success drops −4 pp over 5 steps; under mismatch (β_{true}=1, $\hat{\beta}$=2) the same forced observation costs −11 pp, confirming that passive inference is not merely less useful but actively harmful (Figure 7, Appendix G).

Surprise-tempered updates mitigate mismatch. Table 8 (Appendix G) shows AskOrAct-ST incurs no harm under matched models (+0.0 ± 1.6 pp) and gains +3–+5 pp under mismatch. Three prior decision-layer interventions all failed (−1 to −5 pp), indicating the bottleneck is posterior quality, not the ask-or-act gate.

Generalization. Layout, prior, noise, grid-size, and room-count variations all preserve a positive AskOrAct–NeverAsk gap of +2–+8 pp at K=4, with details provided in Appendix G.

Passive channel degradation. As the fraction p_{drop} of unobserved principal actions rises from 0 to 1, AskOrAct increases Q/ep from 0.66 to 1.11 at K=4 and its advantage over NeverAsk grows from +4.0 ± 5.3 pp to +23.0 ± 3.4 pp. InfoGainAsk holds a fixed ∼1.5 Q/ep regardless of p_{drop}, confirming only the cost-aware policy adapts (Figure 8, Appendix G).

Sub-additive channel use. Channel ablations (both / passive-only / active-only / neither) give interaction I=−26 ± 3.3 pp at K=3 and −20 ± 3.2 pp at K=4: AskOrAct extracts overlapping information from the two channels (Figure 9, Appendix G). Clarification is most valuable when passive inference is weak.

Ablations (see Appendix H for full table) confirm that c_q>0 is the primary driver of selective asking: setting c_q=0 causes over-asking (1.61 q/ep) and degrades success, while the step-counting convention is negligible once cost is non-zero. AskOrAct's first question reduces posterior entropy similarly to InfoGainAsk at K=3, 4, while asking roughly half as often.

6 Discussion

Sub-additive channel use. Channel ablations of AskOrAct show a negative interaction (I= − 26 ± 3 pp at K=3), indicating the policy extracts overlapping information from the two channels. Asking matters most when passive inference is structurally unable to resolve ambiguity, as shown by the +22.6 pp gain in the aliased regime.

The β inverted-U and its mechanism. The non-monotonic assistability finding follows from first-action informativeness: when all candidate goals share the same optimal first action, $I(\text{goal}; a_1) \to 0$ as $\beta \to \infty$ (proof in Appendix I). Increasing β then narrows the action distribution without increasing discriminative power. AskOrAct detects this regime and compensates by asking more. This interaction is invisible in single-channel benchmarks [4, 8, 12] and recognition-only settings [7, 14].

Passive inference as a double-edged sword. Under model mismatch, each forced observation step costs −2.2 pp, which is 2.8× the matched rate, actively poisoning the posterior as shown in Figure 7. Straightforward passive-aware heuristics such as overlap checks and one-step lookahead fail to improve over AskOrAct because the myopic cost comparison already internalises much of the passive information value. An optimal policy should gate not only *whether* to ask, but *when* to stop trusting passive evidence—a design consideration absent from current ask-or-act methods.

Limitations and Future Work. The question bank is fixed at 3 types; open-ended questions require a generative answer model. Scaling to larger environments would require particle-filter or amortised inference; real principals may be non-stationary. The β inverted-U and observation-hurts findings are regime-dependent, and quantitative thresholds are benchmark-specific. Future work could investigate failure-aware objectives (the myopic surrogate does not price wrong picks) and adaptive c_q.

7 Conclusion

We presented a controlled benchmark for ask-or-act decisions with both a diagnostic and a constructive contribution. On the diagnostic side, AskOrAct uses the two channels sub-additively (I= − 26 pp), assistability is non-monotonic in principal rationality, and under mismatch observation actively poisons the posterior at −2.2 pp per step. On the constructive side, surprise-tempered posterior updates in AskOrAct-ST mitigate mismatch with no harm when matched and +1–+5 pp gains under mismatch, after three decision-layer interventions all failed. This suggests posterior quality is at least as important as the ask-or-act policy under misspecification. As a scope note, these findings characterise behaviour in a

controlled gridworld under a Boltzmann principal, and generalization to richer environments and real human principals remains an open question. Code and data are available at https://github.com/Yufannnn/AskOrAct.

AI Tool Declaration

I used Claude Sonnet 4.6 (Anthropic) for coding assistance, debugging, and LaTeX editing throughout the project. I am responsible for all research decisions, experimental design, and the content of this report.

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A Prior Work Comparison

Table 2: Illustrative comparison to closest prior work (not an exhaustive survey). ✓ = present; ∘ = partial; – = absent.

Feature	CLIPS	GRD	PAGR	ATD	ShootFirst	**Ours**
Action-based goal inference	✓	∘	✓	–	–	✓
Explicit ask-or-act decision	–	–	∘	✓	✓	✓
Typed questions with cost c_q	–	–	–	✓	–	✓
Task completion (not just recog.)	✓	–	–	✓	✓	✓
Controlled ambiguity-K sweep	–	–	–	–	–	✓
Ground-truth task success	∘	–	–	–	∘	✓

B Derivation of the Ask-or-Act Decision Rule

The *AskOrAct* decision is derived from Bayes-risk minimisation under a team-cost objective. Let b_t be the posterior over candidate goals $\mathcal{G}$ at step t, and let $d^*(s, g)$ denote the obstacle-aware shortest-path distance computed via A* from state s to goal g.

Cost of acting. If the assistant acts now, the expected cost is

$$\mathrm{CostAct}(b_t, s_t) = \sum_{g \in \mathcal{G}} b_t(g)\, d^*(s_t, g).$$

Cost of asking question q. Asking consumes one step costing 1, incurs a question penalty c_q on the team cost, and produces an answer α drawn from the noisy channel $P(\alpha \mid q, g)$. The posterior after observing answer α is $b_{t,q,\alpha}(g) \propto b_t(g)\, P(\alpha \mid q, g)$. The assistant stays in place while asking, so CostAct is evaluated at s_t:

$$\begin{aligned}\mathrm{CostAsk}(q, b_t, s_t) &= 1 + c_q \\ &\quad + \textstyle\sum_\alpha P(\alpha \mid q, b_t)\, \mathrm{CostAct}(b_{t,q,\alpha}, s_t),\end{aligned}$$

where $P(\alpha \mid q, b_t) = \sum_g b_t(g)\, P(\alpha \mid q, g)$ is the predictive distribution over answers.

Decision rule. The policy selects

$$\pi(b_t, s_t) = \begin{cases} \mathrm{ask}(q^*) & \text{if } \min_q C_{\mathrm{Ask}}(q) < C_{\mathrm{Act}}(b_t), \\ \mathrm{act}(\hat{g}) & \text{otherwise,} \end{cases}$$

where C_{Ask} and C_{Act} are shorthand for *CostAsk* and *CostAct*, the ask is subject to the entropy gate / window / budget constraints,

and $\hat{g} = \arg\max_g b_t(g)$ is the MAP goal. This rule is myopic in that it does not plan a sequence of questions but rather performs a one-step lookahead comparing immediate action against the best single question. This simplification makes the decision $O(|\mathcal{Q}| \cdot |\mathcal{G}| \cdot |\mathcal{A}_{\text{ans}}|)$ per step, which is exact and efficient for the small question banks and candidate sets studied here.

C Within-Template T3 Control and Full Results

Table 3: Within-template T3 (color-only). Isolates the effect of K by holding template constant.

K	Policy	Success	Q/ep
2	AskOrAct	95%	0.61
	NeverAsk	97%	0.00
3	AskOrAct	90%	0.67
	NeverAsk	87%	0.00
4	AskOrAct	92%	0.61
	NeverAsk	85%	0.00

Table 4 provides per-(K, ε) success rates and questions asked for the five core policies, averaged over $\beta \in \{1, 2, 4\}$ and 5 replicate seeds. Entries at K=1 are omitted since all policies achieve 100% success with 0 questions.

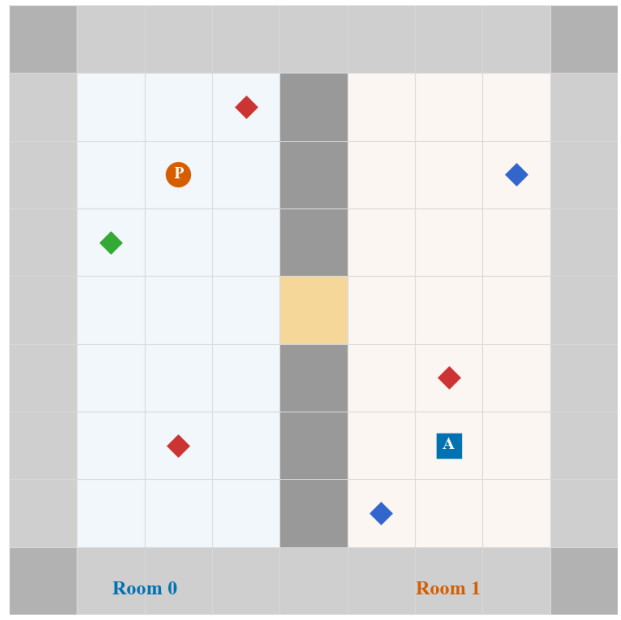

Figure 3: Two-room 9×9 gridworld. Room 0 (blue, left) and Room 1 (yellow, right) are separated by a mid-wall with a single doorway. Gold rings mark the K=2 candidate goals; P = principal, A = assistant.

D Hyperparameter Sensitivity

The three key *AskOrAct* hyperparameters are the question cost c_q, the entropy gate τ_H, and the ask window. Their roles and approximate sensitivities are as follows.

Question cost c_q. The ablation in Table 13 confirms this is the primary driver, as c_q=0 leads to over-asking and performance collapse. For $c_q \in [0.3, 1.0]$, the policy behaviour is qualitatively similar and selectivity increases with c_q, though success is non-monotone, with an optimum around c_q=0.5.

Entropy gate τ_H. When $H(b_t) \leq \tau_H$, the posterior is concentrated enough that asking yields negligible information gain. For $\tau_H \in [0.2, 0.5]$, the policy is largely insensitive because the explicit IG comparison in CostAsk already discourages low-value questions, and the entropy gate simply provides a fast early-exit.

Ask window. Restricting asking to the first 6 steps prevents late questions that cannot pay off before the step budget is exhausted. Extending the window to 10 shows diminishing returns at K=3, 4 because the posterior has usually converged by step 6 via action observations.

E Episode Generation, Templates, and Hyperparameters

Episode generation procedure. Each episode is generated from a single integer seed. The procedure, implemented in `src/world/worldgen.py`, is as follows.

1. Build a 9×9 grid with outer walls and a vertical mid-wall at column 4; place the doorway at (4, 4).
2. Sample a template $t \in \{0, 1, 2, 3\}$ and a true goal colour/type/room uniformly at random.
3. Generate K *candidate* objects: all share the attribute(s) that the template exposes. For $K \geq 2$ and the room template t=2, the constraint forces all candidates to the same room, so this template is excluded when $K \geq 2$ in the two-room setting.
4. Generate $M-K$ *distractor* objects with rejection sampling to ensure they do not match the instruction.
5. Sample positions for the K candidates with a minimum pairwise grid distance of $\lfloor N/2 \rfloor$, and for $K \geq 2$ at least one candidate is required in each room.
6. Place distractors uniformly in remaining free cells and shuffle object IDs. Place principal and assistant at randomly sampled free cells (distinct from each other and from objects).
7. Emit the instruction by formatting the selected template with the true goal's attributes.

Up to 200 retries are performed with incremented seeds if any constraint check fails; a `RuntimeError` is raised if all retries fail.

Instruction templates. Four surface-form templates are used as shown in Table 6. Ambiguity level K determines how many objects share the instruction. The color-only template T3 supports all K values tested from K=1 to 4, while the color+type template T1 allows at most 1 match and is used only at K=1.

Initialisation and answer model. The prior is uniform over candidates: $b_0(g) = 1/K$ for each candidate, 0 for distractors. Three question types are available, addressing color, type,

Table 4: Per-(K, ε) results for the five core policies, averaged over β. K=1 omitted (trivial: all 100%).

		AskOrAct		InfoGainAsk		AlwaysAsk		NeverAsk		RandomAsk	
K	ε	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep
2	0.00	97.3%	0.60	97.2%	0.99	94.8%	1.83	91.5%	0.00	92.1%	0.60
2	0.05	96.8%	0.65	97.0%	1.05	94.6%	1.87	91.3%	0.00	91.8%	0.63
2	0.10	96.0%	0.69	96.4%	1.11	94.5%	1.91	90.5%	0.00	91.2%	0.67
3	0.00	92.5%	0.63	95.4%	1.17	91.0%	2.24	84.8%	0.00	86.0%	0.79
3	0.05	91.8%	0.67	94.8%	1.21	90.2%	2.29	83.8%	0.00	85.5%	0.82
3	0.10	91.0%	0.72	94.3%	1.26	89.5%	2.34	82.8%	0.00	84.8%	0.85
4	0.00	88.2%	0.57	92.5%	1.26	87.1%	2.52	79.8%	0.00	80.9%	0.87
4	0.05	87.3%	0.61	91.7%	1.31	86.4%	2.58	78.9%	0.00	80.2%	0.91
4	0.10	86.4%	0.64	91.0%	1.36	85.7%	2.63	78.0%	0.00	79.5%	0.94

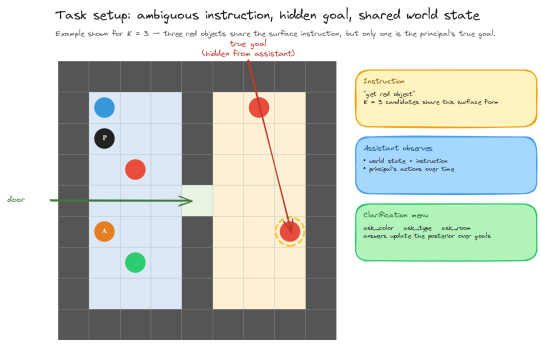

Figure 4: Task setup. The principal has a hidden goal; the assistant sees the world state, an ambiguous instruction, and the principal's movement. Questions collapse the posterior over candidate goals.

and room. Answers follow a noisy channel: $P(\alpha \mid q, g) = (1-\eta)\,\mathbf{1}[\alpha{=}\text{true}(g,q)] + \eta/(|\mathcal{A}_q|-1)\,\mathbf{1}[\alpha{\neq}\text{true}(g,q)]$, with η=0.05 (color), 0.10 (room), 0.20 (type).

F Policy Pseudocode

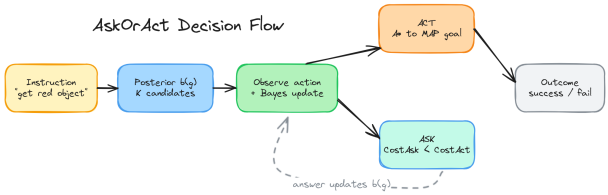

Figure 5: AskOrAct decision flow: instruction → posterior → observe + update → ASK or ACT. Dashed: answer feeds back.

Algorithm 6 gives pseudocode for the *AskOrAct* policy. All baselines share the same act branch of greedy A* toward the MAP goal and differ only in their ask-gating condition.

Table 5: Environment and policy hyperparameters.

Parameter	Value
Grid size	9×9, two rooms
Objects	6 (3 types × 3 colours)
Step budget $T_{\max}$	200
Question bank $\lvert Q\rvert$	3 (color, room, object)
Ask budget $Q_{\max}$	3
Ask window	steps ≤ 6
Question cost c_q	0.5
Entropy gate τ_H (AskOrAct)	0.3
IG threshold τ_{IG}	0.01
Answer noise (color/room/object)	0.05 / 0.10 / 0.20
Principal rationality β	2.0 (default)
Principal noise ε	0.05 (default)

Table 6: Instruction templates. Object attributes: type ∈ {key, gem, coin}, color ∈ {red, blue, green}, room ∈ {room0, room1}.

ID	Surface form	Ambiguity (shared attr.)
T0	"get {type}"	type only
T1	"get {color} {type}"	color + type
T2	"get {type} in {room}"	type + room
T3	"get {color} object"	color only

POMCPPlanner uses a 100-particle belief with UCT exploration constant c=1.4 and random rollouts scoring −1 per step and +10 for a correct pick, with the tree rebuilt each step.

G Robustness Sweep Details

Answer noise sweep. Table 9 shows AskOrAct vs. NeverAsk success at K=3 and K=4 across global noise levels $\eta \in \{0.0, 0.05, 0.1, 0.15, 0.2\}$ (overriding per-type defaults).

At K=3, noise = 0.1 is the hardest setting because noisy answers briefly mislead the posterior, reversing the advantage. The advantage re-opens at noise = 0.2 because NeverAsk is

Require: belief b_t, state s_t, asked set A, step t

$g^* \leftarrow \arg\max_g b_t(g)$ ▹ MAP goal
$C_{\text{act}} \leftarrow \sum_g b_t(g)\, d^*(s_t, g)$
if $H(b_t) > \tau_H$ **and** $|A| < Q_{\max}$ **and** $t \le W$ **then**
  for each $q \in \mathcal{Q} \setminus A$ **do**
    $C_q \leftarrow 1 + c_q + \sum_\alpha P(\alpha|q, b_t)\, C_{\text{act}}(b_{t,q,\alpha}, s_t)$
  end for
  if $\min_q C_q < C_{\text{act}}$ **then**
    return ask($\arg\min_q C_q$)
  end if
end if
return act(A*-step toward g^*)

Figure 6: AskOrAct (τ_H=0.3, $Q_{\max}$=3, W=6, c_q=0.5).

Table 7: Baseline ask-gating conditions (act branch is always A* to MAP goal).

Policy	Ask condition
NeverAsk	Never
AlwaysAsk	$\lvert A\rvert < Q_{\max}$ and $H(b_t) > \tau_H$
InfoGainAsk	AlwaysAsk and $\text{IG}(q^*, b_t) > \tau_{\text{IG}}$
EasyInfoGainAsk	$\lvert A\rvert < 1$ and $\text{IG}(q^*, b_t) > \tau_{\text{IG}}$
RandomAsk	InfoGainAsk with $q \sim \text{Uniform}(\mathcal{Q} \setminus A)$
POMCPPlanner	POMCP [10] tree search, 250 iter, horizon 8

equally hurt by high noise. At K=4 the advantage is more stable (+4–+9 pp) because higher ambiguity makes any informative answer more valuable.

Principal model mismatch. The assistant uses fixed $\hat{\beta}$=2 while the principal's true β is varied. Table 10 shows results at K=3 averaged over ε.

The largest gain from asking of +13.5 pp occurs at the hardest mismatch of β_{true}=1 with a near-random principal, where action observations provide little information and direct questions become the only reliable disambiguator. At β_{true}=4 the principal is highly rational and easy to model, so the non-asking baseline already performs well.

Scale-K evaluation. Table 11 extends results to $K \in \{5, 6\}$ using the same hyperparameters and 100 episodes per condition.

Both asking policies degrade gracefully, while NeverAsk falls substantially. The AskOrAct–NeverAsk gap grows to +11.6 pp at K=5 and +14.4 pp at K=6, consistent with value-of-information being most valuable at high ambiguity.

Path overlap stratification. Define *overlap* = $|\{g \in \mathcal{G} : a^*(g) = \text{mode}\}|/K$, the fraction of candidates sharing the modal optimal first action. Table 12 stratifies K=3 episodes by this metric. The "diverge" bucket is small (n=7), so the pattern is suggestive rather than definitive.

H Ablations

Table 13 averages over ablation variants including wrong-pick termination (episode ends on wrong pick vs. continues)

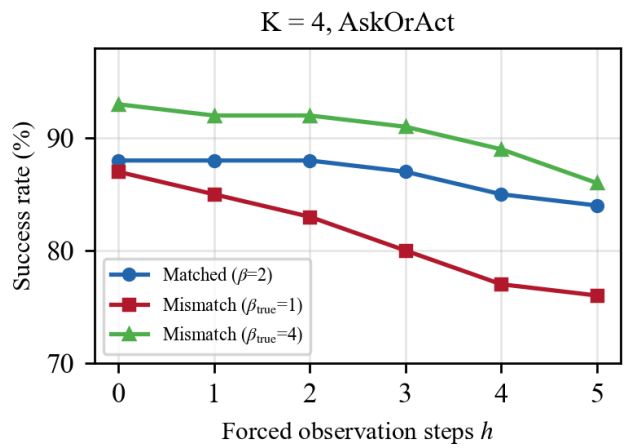

Figure 7: Forced observation horizon h vs. success at K=4. Under mismatch (β_{true}=1), more observation actively hurts.

Table 8: AskOrAct vs. AskOrAct-ST (surprise-tempered). Δ: gain from tempering.

K	Condition	AskOrAct	ST	Δ (pp)
3	Matched (β=2)	95%	95%	+0.0 ± 1.6
3	Mismatch (β=1)	78%	81%	+3.0 ± 2.5
3	Mismatch (β=0.5)	73%	76%	+3.0 ± 2.0
4	Matched (β=2)	82%	82%	+0.0 ± 0.0
4	Mismatch (β=1)	71%	72%	+1.0 ± 1.9
4	Mismatch (β=0.5)	54%	59%	+5.0 ± 2.7

and deadline margins (±2 steps around optimal). Mode A with c_q=0 over-asks at 1.61 q/ep and degrades success, while Modes B and C show that the step-counting convention has negligible effect once c_q>0. Non-zero question cost is the primary driver of selective asking. A sensitivity sweep over $\tau_H \in [0.2, 0.5]$ and ask window $\in \{6, 10\}$ shows negligible effect once c_q>0.

I Formal Analysis of the β Inverted-U

Observation. When candidate goals share the same optimal first action (first-action aliasing), $I(\text{goal}; a_1) \to 0$ as $\beta \to \infty$.

Proof sketch. Let $a^*(g)$ denote the optimal first action for goal g. Suppose $a^*(g) = a^*$ for all $g \in \mathcal{G}$. Under the Boltzmann policy, as $\beta \to \infty$ the action distribution concentrates on a^* regardless of g: $P(a \mid g) \to \mathbf{1}[a = a^*]$ for all g. The marginal $P(a)$ then equals $P(a \mid g)$, giving $I(\text{goal}; a_1) = 0$. At finite β, sub-optimal actions may still carry residual goal information, but this can decrease with β in the aliased case—contrary to the intuition that "more rational = more informative." Increasing β therefore narrows the action distribution without increasing discriminative power, while reducing the sub-optimal-action diversity that could accidentally reveal intent.

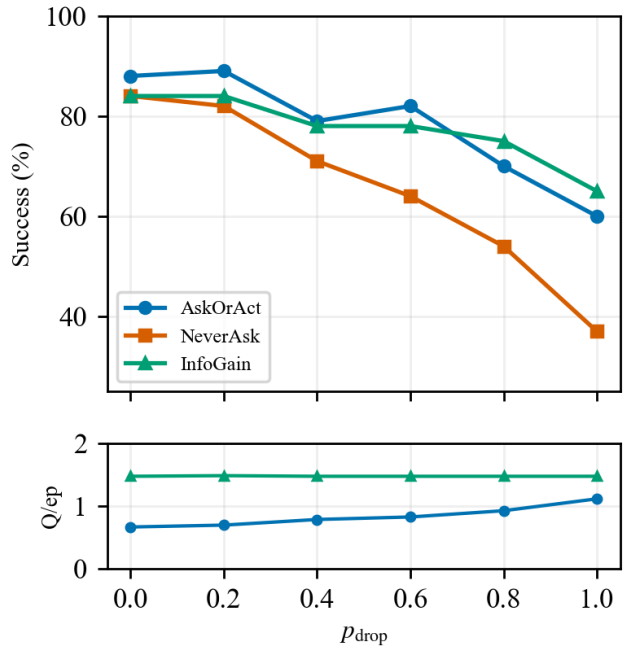

Figure 8: Passive channel degradation at K=4. Top: success rate vs. p_{drop}. Bottom: Q/ep. AskOrAct adaptively asks more as passive inference weakens.

Table 9: Success rate (%) under global answer noise, averaged over (ε, β) conditions.

	Noise level η				
Policy	0.0	0.05	0.10	0.15	0.20
K=3					
AskOrAct	93.0	92.2	92.0	92.5	95.0
NeverAsk	84.8	83.8	94.0	89.0	94.0
K=4					
AskOrAct	89.5	88.0	86.5	86.0	85.5
NeverAsk	79.8	78.9	80.5	79.5	81.5

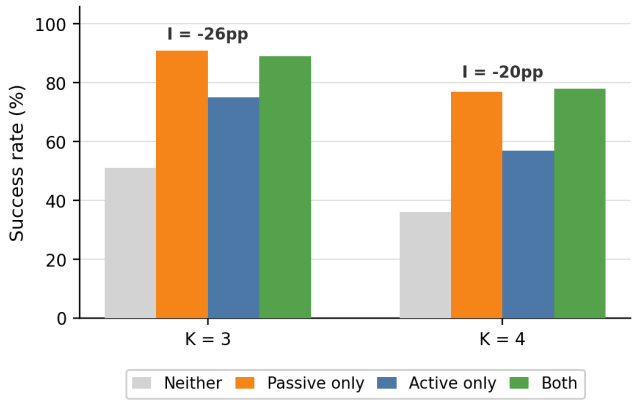

Figure 9: Channel ablation interaction under AskOrAct. Negative = sub-additive channel use. I values annotated per K.

Table 10: Success rate (%) under principal model mismatch ($\hat{\beta}$=2 fixed, K=3).

Policy	β_{true}=1	β_{true}=2	β_{true}=4
AskOrAct	83.5	91.7	95.0
NeverAsk	70.0	83.8	92.5
Δ	+13.5	+7.9	+2.5

Table 11: Success rate (%) at $K \in \{5, 6\}$, averaged over (ε, β).

Policy	K=5	K=6
AskOrAct	82.1	76.8
InfoGainAsk	87.5	83.2
NeverAsk	70.5	62.4

Table 12: Path overlap vs. success at K=3.

Overlap	AskOrAct	NeverAsk	Δ	n
Diverge (<0.5)	100.0%	100.0%	0	7
Partial (0.5–0.99)	95.2%	98.4%	−3.2	62
Aliased (1.0)	87.1%	64.5%	+22.6	31

Table 13: Ablation results for *AskOrAct* at K=3, 4, averaged over (ε, β) and ablation variants (wrong-pick-fail ∈ {T,F}, deadline margin ∈ {3, 5, 7}). Lower success than Table 1 reflects the harsh variant mix.

Mode	c_q	Ask-step	Success	Q/ep
A	0.0	yes	69.2%	1.61
B	0.5	no	73.6%	0.64
C	0.5	yes	73.8%	0.65